

Chapter 15. Network Infrastructure Device Types

15.1. Introduction

Matter aims to build a universal IPv6-based communication protocol for smart home devices, and in principle almost any IPv6-bearing Wi-Fi, Thread, or Ethernet network is suitable for Matter deployment (see *Network Topology* in [\[MatterCore\]](#)).

The Network Infrastructure device types defined in this chapter are designed to provide a reliable, secure, and performant connectivity experience for Matter devices and their users. To achieve this, these device types expose Matter application clusters for the purposes of network management, diagnostics, and configuration. They also include requirements on lower layers (including the network and physical layers) that contribute to the desired connectivity experience.

Note that Matter devices and applications cannot rely on the presence of Matter-certified Network Infrastructure Devices, or the presence of network features mandated by these device types, for their correct operation.

15.2. Common Requirements

15.2.1. Minimum Number of Devices to Support

The count of IoT devices within a smart home can quickly grow to a substantial number when factoring in the quantity of light points, light switches, and an array of sensor devices that all require connectivity. Consequently, the network infrastructure must be capable of accommodating a substantial number of devices to avoid hitting constraints when users wish to add "one more" product. The requirements in this chapter were designed to result in an IP network (which MAY be a combination of Ethernet LAN, Wi-Fi WLAN, and Thread PAN networks) with a minimum of 300 Matter devices.

- Requirements related to Wi-Fi Access Points and Thread Border Routers are detailed in the sections for the [Network Infrastructure Manager](#) and [Thread Border Router](#) device types below.

15.3. Network Infrastructure Manager Device Type

A Network Infrastructure Manager provides interfaces that allow for the management of the Wi-Fi, Thread, and Ethernet networks underlying a Matter deployment, realizing the *Star Network Topology* described in [\[MatterCore\]](#).

Examples of physical devices that implement the Matter Network Infrastructure Manager device type include Wi-Fi gateway routers.

Relevant hardware and software requirements for Network Infrastructure Manager devices are defined in [Section 15.3.6, "Other Requirements"](#) and within the clusters mandated by this device type.